We are going to look at different types of functions.

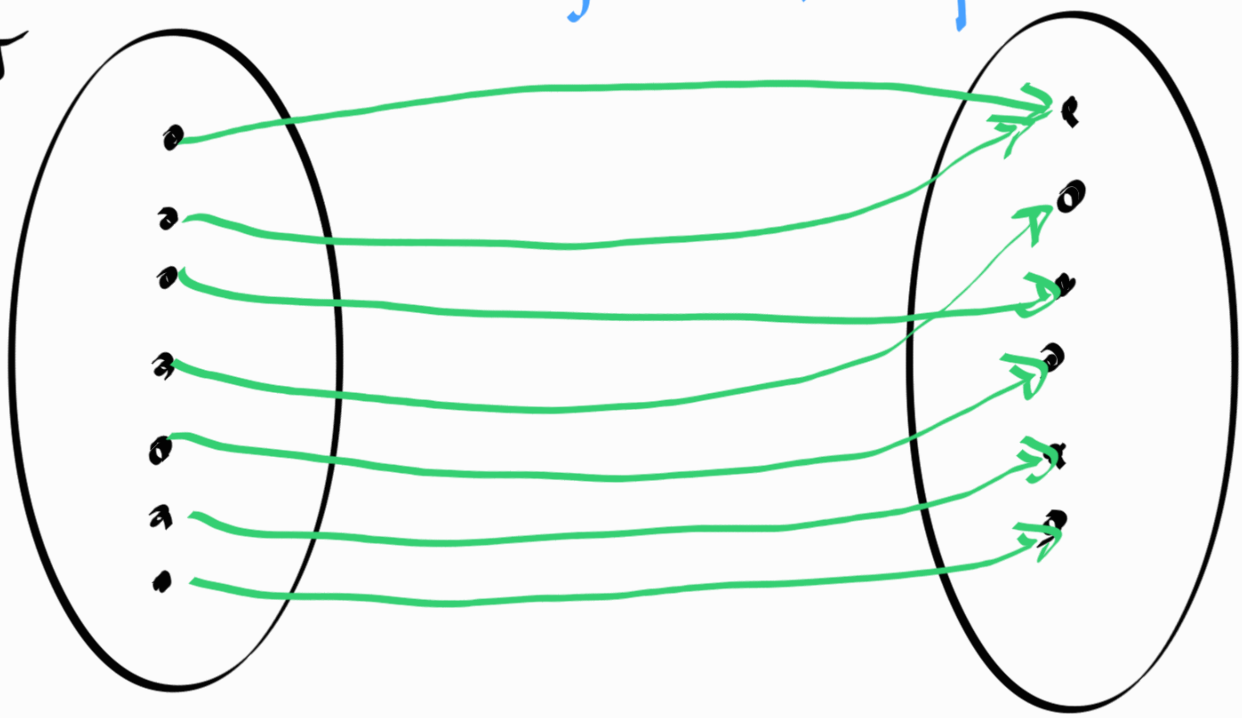
<u>Def</u> onto/surjective functions

A function $f: A \to B$ is called onto (surjective) iff $[\forall b \in B : [\exists a \in A : f(a) = b]]$

$\equiv \forall b \in B$, something in A maps to it.

B

A



Domain

Codomain

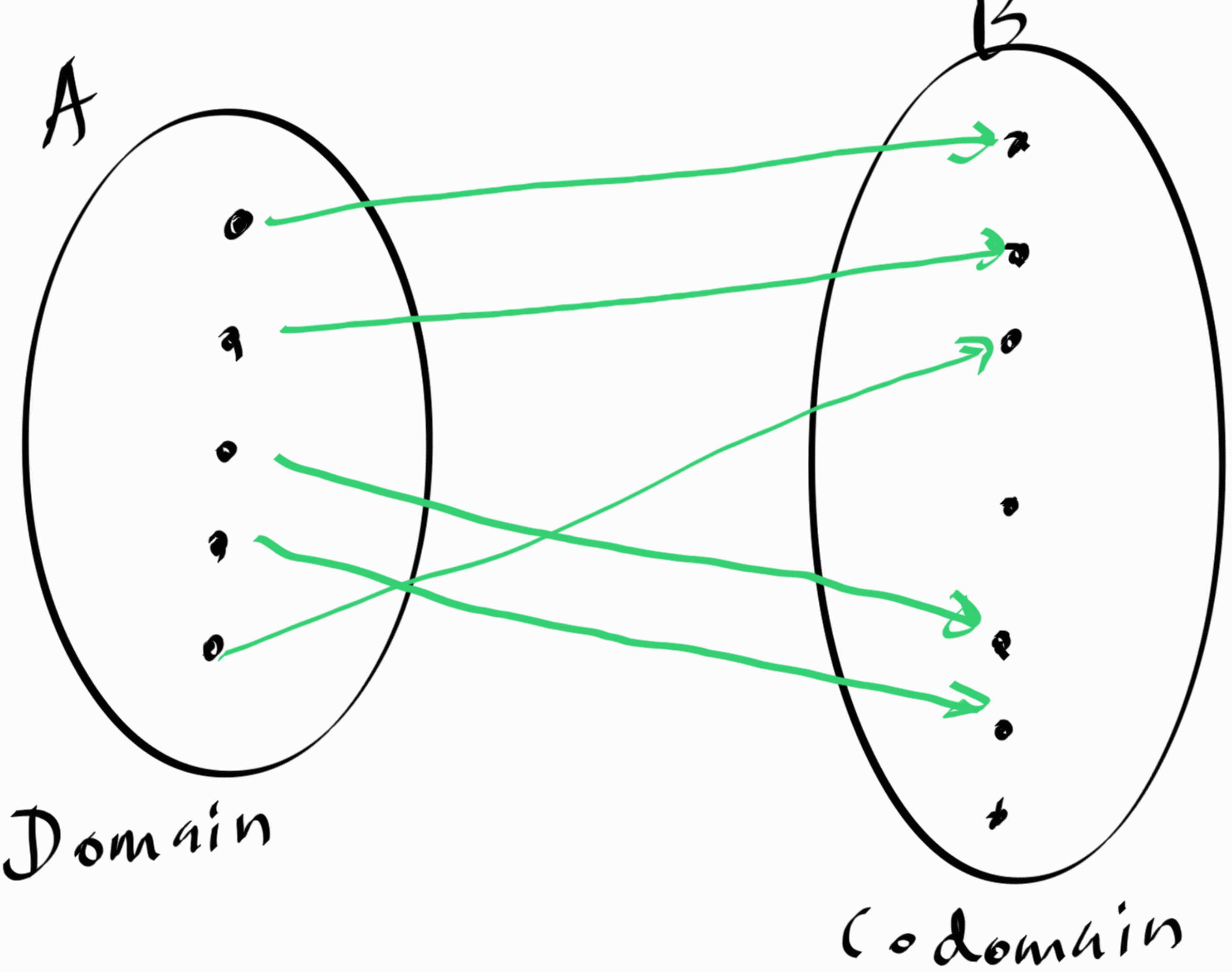
Range = Codomain

Def one-to-one / injective functions.

A function $f: A \to B$ is called one-to-one (injective)

iff $[\forall a_1, a_2 \in A: a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)]$

$\equiv \forall b \in B$, at most 1 element in A maps to it.

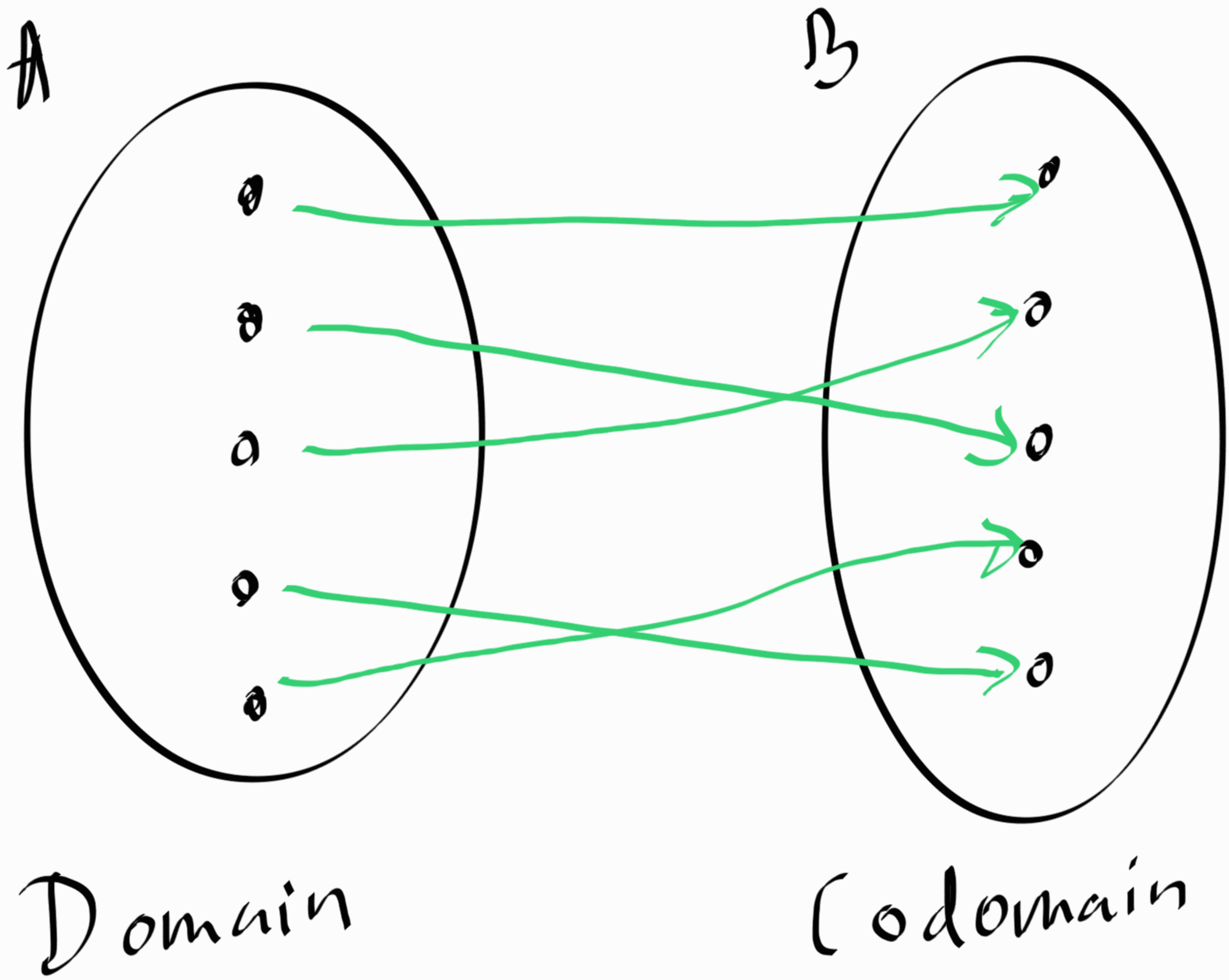


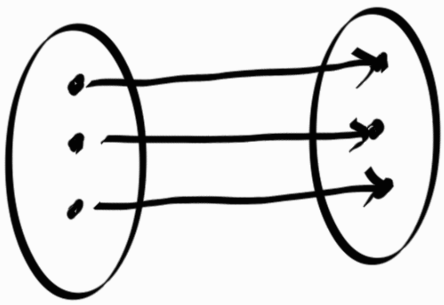
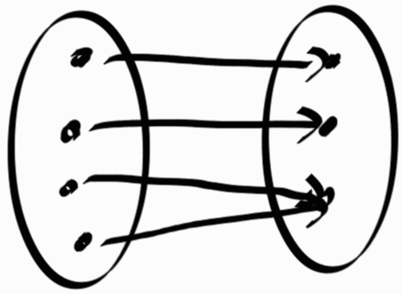
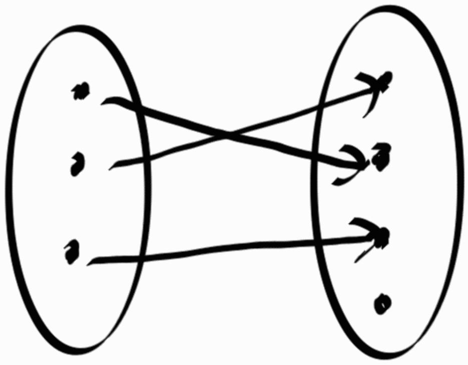
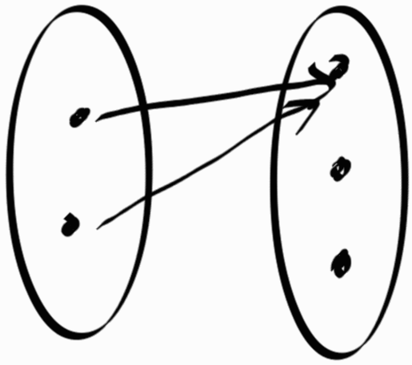
Injective and surjective are **NOT** the opposite of each other.

Def Bijection

A function $f: A \to B$ is called a bijection if f is onto and one-to-one.

$\equiv \forall b \in B$, exactly 1 element in A maps to it.


A
B
Domain
Codomain

	1:1 (one-to-one)	not 1:1
onto		
not onto		

How can we prove a function f is onto or 1:1 ?

onto

WTS

$$\left[\forall b \in B : \left[\exists a \in A : f(a) = b \right] \right]$$

$$\equiv$$

$$b \in B \Rightarrow \left[\exists a \in A : f(a) = b \right]$$

example:

$$S : \mathbb{Z} \longrightarrow \mathbb{Z} \quad , \quad S(x) = x + 1$$

Let's prove that s is an onto function

WTS: $b \in \mathbb{Z} \Rightarrow \left[\exists a \in \mathbb{Z} : S(a) = b \right]$

Proof

$b \in \mathbb{Z}$ by assumption

Let's construct $a \in \mathbb{Z}$ s.t $S(a) = b$

Consider
$a = b - 1$, $a \in \mathbb{Z}$ because $b \in \mathbb{Z}$

now, $S(a) = b - 1 + 1$ by def of S.

$S(a) = b$

So, we found $a \in \mathbb{Z}$ that maps to b.

$\therefore$ S is an onto function. $\blacksquare$

How to show a function $f: A \to B$ is not a onto function.

A function f is onto $\iff$ $[\forall b \in B : \exists a \in A : f(a) = b]$

$a \iff b$

$\neg a \iff \neg b$

$\neg [\forall b \in B : [\exists a \in A : f(a) = b]]$

Note that:

$\neg [\forall x \in S : P(x)] = [\exists x \in S : \neg P(x)]$

$[\exists b \in B : \neg [\exists a \in A : f(a) = b]]$

Note that

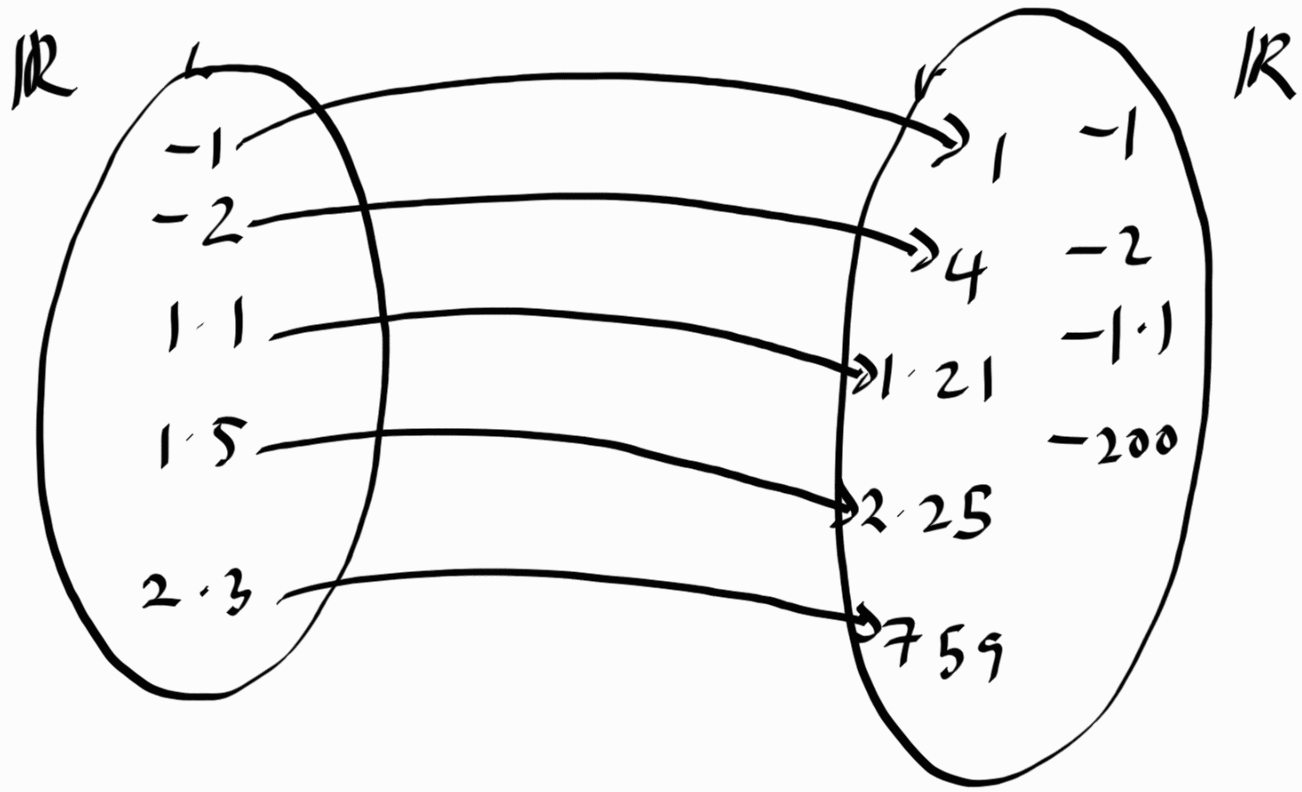
$\neg [\exists x \in S : P(x)] = [\forall x \in S : \neg P(x)]$

$$\left[\exists b \in B : \left[\forall a \in A : \neg(f(a) = b)\right]\right]$$

$$\left[\exists b \in B : \left[\forall a \in A : f(a) \neq b\right]\right]$$

Prove: $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

is not a onto function



Let $b = -1$, $b \in \mathbb{R}$

WTS $\left[\forall a \in \mathbb{R} : f(a) \neq b\right]$

$\forall a \in A: f(a) = a^2$ by def. of f

$\forall a \in A: a^2 \geq 0$ by the property of squares.

$\forall a \in A: a^2 \neq b$ $a^2 \geq 0$ and $b = -1$

$\forall a \in A: f(a) \neq b$ by substitution

we proved an existence of an element $b \in \mathbb{R}$, s.t $[\forall a \in A: f(a) \neq b]$

$\therefore$ function f is <u>not onto</u>.

How to prove a function f is <u>not</u> $1:1$.

Let $f: A \longrightarrow B$, then

function f is $1:1 \iff$

$$\left[\forall a_1, a_2 \in A : a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)\right]$$

$$\equiv$$

function f is not $1:1 \iff$

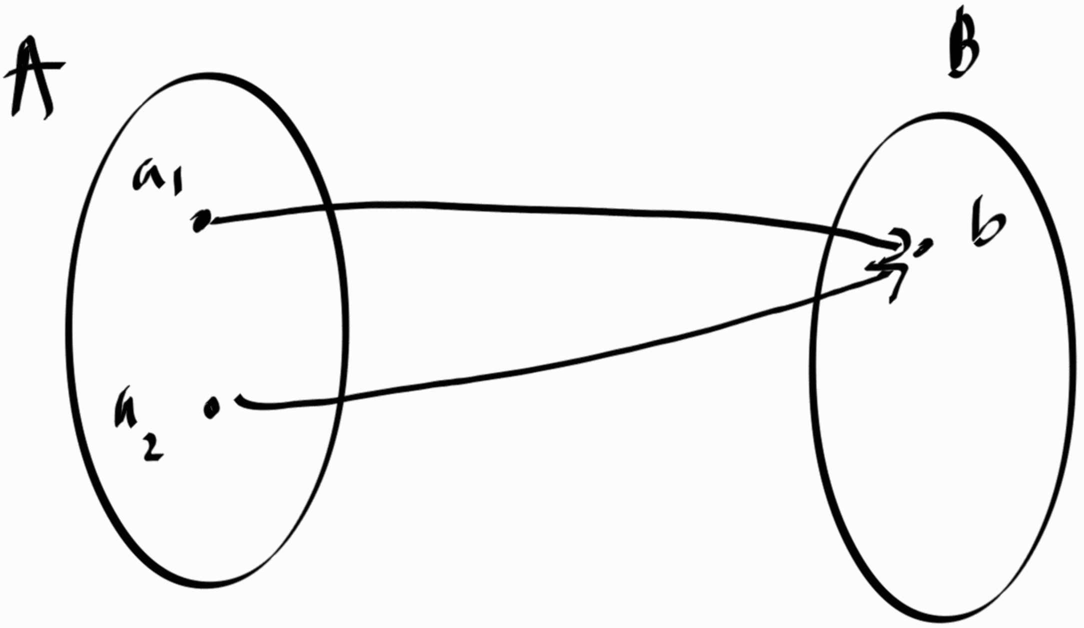
$$\neg\left[\forall a_1, a_2 \in A : a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)\right]$$

$$\neg\left[\forall a_1, a_2 \in A : a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)\right] \equiv$$

$$\left[\exists a_1, a_2 \in A : \neg\left[a_1 \neq a_2 \Rightarrow f(a_1) \neq f(a_2)\right]\right]$$

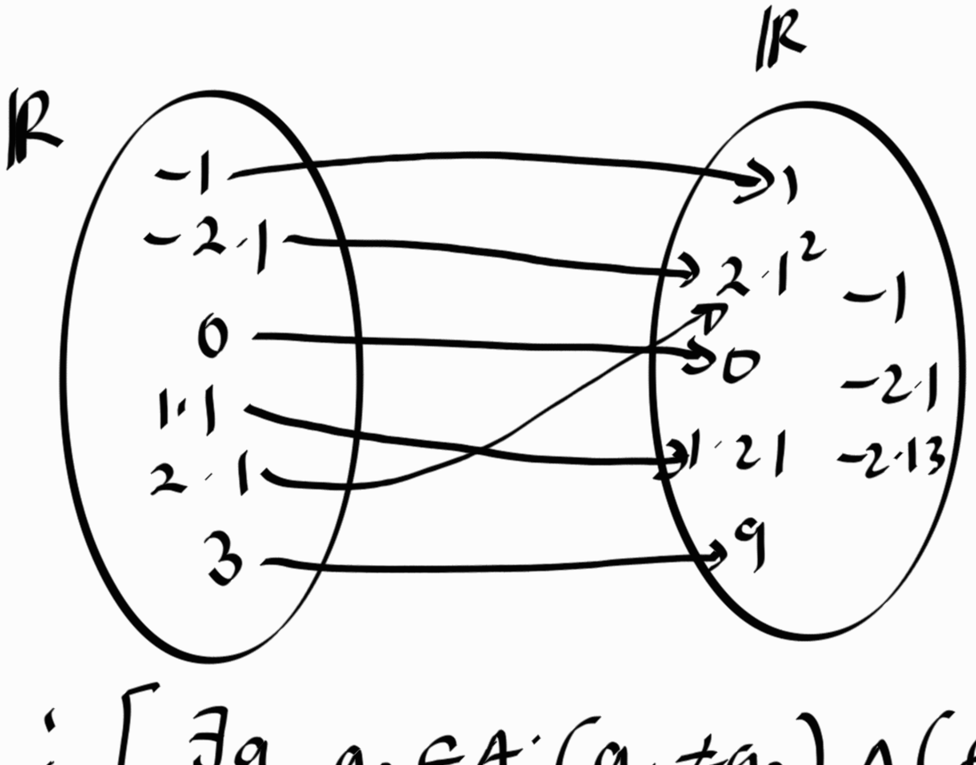
$$\left[\exists a_1, a_2 \in A : (a_1 \neq a_2) \wedge (f(a_1) = f(a_2))\right]$$

$$\neg(P \Rightarrow q) \equiv (P \wedge \neg q)$$



Prove: $f: \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$

is not 1:1.



WTS: $[\exists a_1, a_2 \in A : (a_1 \neq a_2) \wedge (f(a_1) = f(a_2))]$

Consider $a_1 = 2.1$ $\quad a_2 = -2.1$

$(a_1 \neq a_2)$ because $a_1 = 2.1$ and $a_2 = -2.1$

$f(a_1) = 2 \cdot 1^2$, $f(a_2) = 2 \cdot 1^2$ by the def of f.

$f(a_1) = f(a_2)$

$(a_1 \neq a_2) \wedge (f(a_1) = f(a_2))$

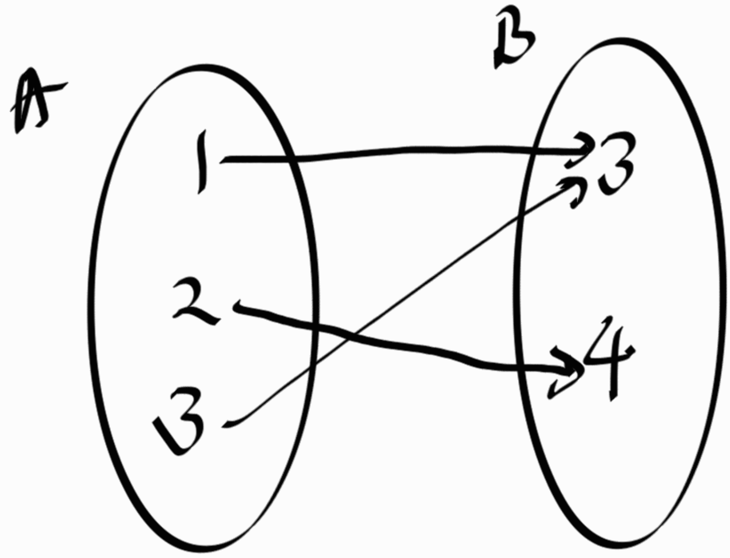
We showed that there exist two distinct elements in the domain $\mathbb{R}$ that is mapped to the same element in codomain $\mathbb{R}$.

$\therefore$ function f is not 1:1 $\blacksquare$

Popup Quiz 06

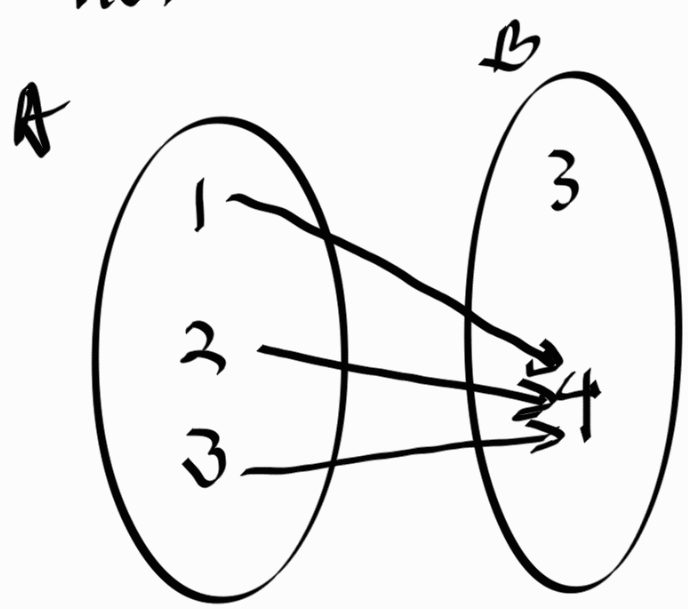
$A = \{1, 2, 3\}$ $\quad$ $B = \{3, 4\}$

1. An onto function $f: A \to B$

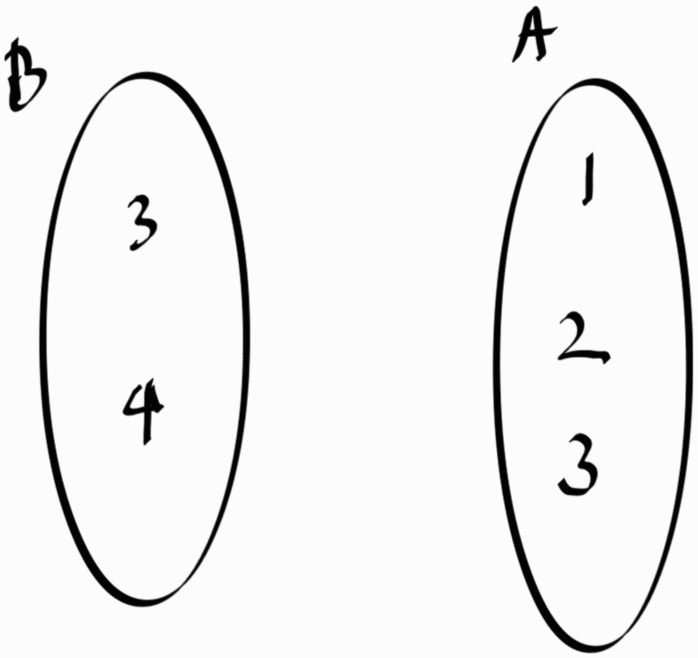


a	$f(a)$
1	3
2	4
3	3

2. A function $g: A \to B$ that is not onto



3. A function $h: B \rightarrow A$ that is onto.



There is no possible onto function h between $B \rightarrow A$